## Unit 4 Progress Check: FRQ

**Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.**

WataDine is one of a city's many restaurants that serve lunch and dinner in a monopolistically competitive market. Assume WataDine, as a typical restaurant in the city, is currently producing the profit-maximizing output level, and earns positive economic profit.

(a) How is monopolistic competition similar to each of the following market structures?

(i) Perfect competition

(ii) Monopoly

(b) WataDine is currently earning positive economic profits. Draw a correctly labeled graph for WataDine in short-run equilibrium and show each of the following.

(i) The profit-maximizing quantity, labeled $Q_M$

(ii) The profit-maximizing price, labeled $P_M$

(c) Given that WataDine is currently earning positive economic profits, what will happen to each of the following in the long run?

(i) WataDine's economic profit. Explain.

(ii) WataDine's demand curve for its restaurant meals.

(d) Assume WataDine is in long-run equilibrium.

(i) Is WataDine taking advantage of its economies of scale? Explain.

(ii) Would WataDine produce the productively efficient output? Explain.

**1.** Respond to all parts of the question.

**Part (a)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 ✓ |
|---|---|---|

The response explains one way in which monopolistic competition is similar to perfect competition (e.g., monopolistic competition is similar to perfect competition because they both have large numbers of firms and easy entry and exit) AND explains one way in which monopolistic competition is similar to a monopoly (e.g., they both have market power, face a

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downward-sloping demand curve, and their price is greater than the marginal cost $P > MC$ at the profit-maximizing quantity).

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

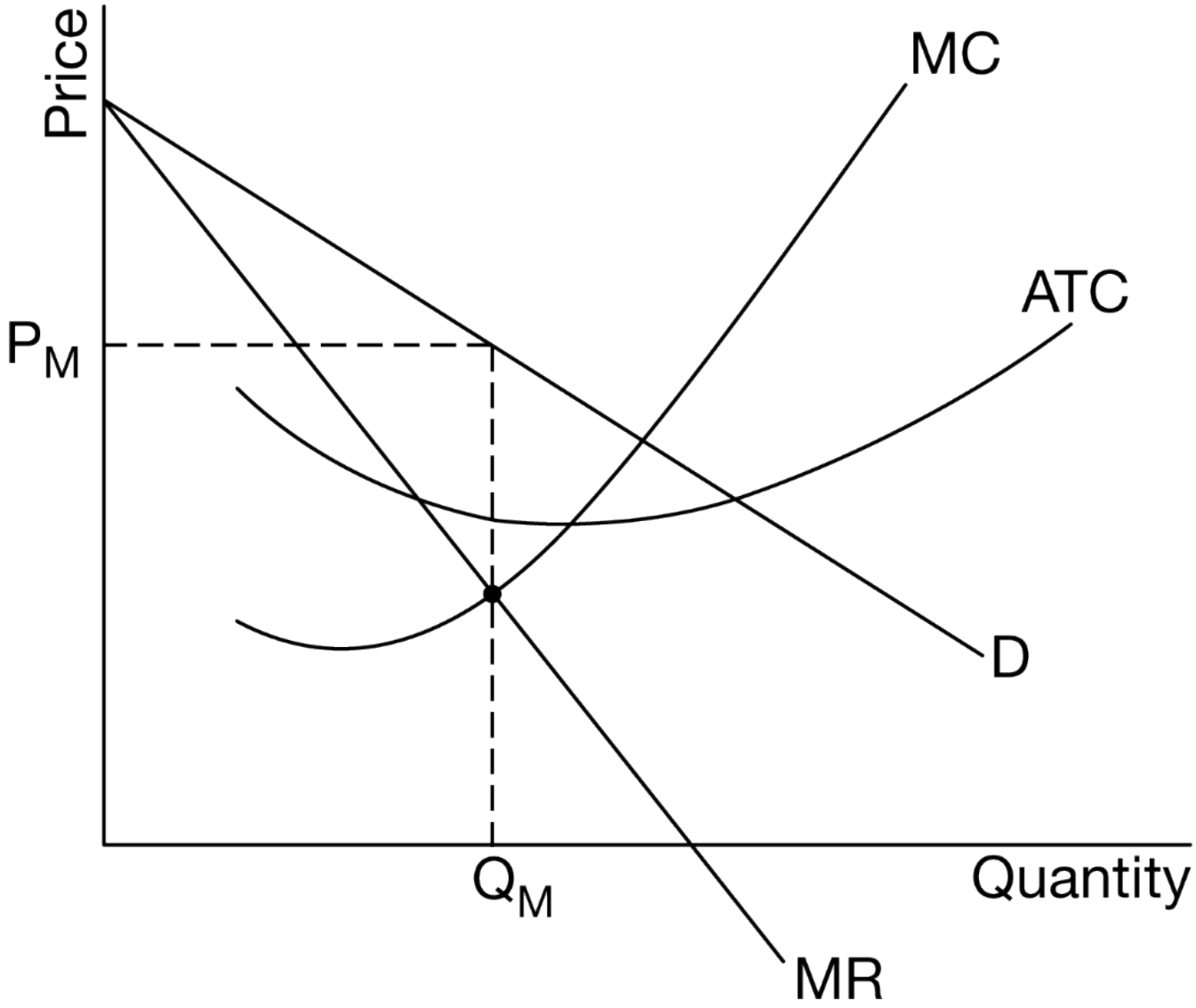
| 0 | 1 | 2 | 3 | 4 ✓ |
|---|---|---|---|---|

The response accurately includes all four of the criteria below..

- ☐ A correctly labeled graph of the firm, with a downward-sloping demand curve and a downward-sloping marginal revenue curve below it.
- ☐ The profit-maximizing quantity labeled $Qm$ at $MR = MC$
- ☐ $Pm$ from the demand curve above $Qm$
- ☐ The $ATC$ curve below the demand curve at $Qm$

Solution

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**Part (c)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 ✓ |
|---|---|---|

The response accurately includes both of the criteria below.

- ☐ States that economic profits will decrease to zero and explains that short-run positive economic profits will attract new firms to enter the market.
- ☐ States that the firm's demand curve will shift to the left.

**Part (d)**

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both state no and explain why in order to receive each point.

| 0 | 1 | 2 ✓ |
|---|---|---|

The response accurately includes both of the criteria below.

- ☐ The response states that no, WataDine is not taking advantage of economies of scale, and explains that the profit-maximizing quantity, $Qm$ , is not produced at the point of minimum average total cost or because marginal cost will exceed the marginal revenue if WataDine increases output beyond $Qm$ ,resulting in a decrease in economic profit.
- ☐ The response states no, the firm would not produce the productively efficient output in long-run equilibrium, and explains that the firm would be operating at a loss, since price will be below $ATC$ , or the firm would be operating at a point where $MC$ is greater than $MR$ .

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**Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.**

(a) Draw a correctly labeled graph for a monopoly and show each of the following.

(i) The profit-maximizing quantity, labeled $Q_M$

(ii) The profit-maximizing price, labeled $P_M$

(b) Assume the monopoly now engages in perfect price discrimination. On your graph in part (b), show the profit-maximizing quantity for the price discriminating monopoly, labeled $Q_{PD}$.

(c) Based on your answer to part (b), will a perfectly competitive market produce a larger, smaller, or the same quantity as the perfectly price discriminating monopoly? Explain.

**2.** Respond to all parts of the question.

**Part (a)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

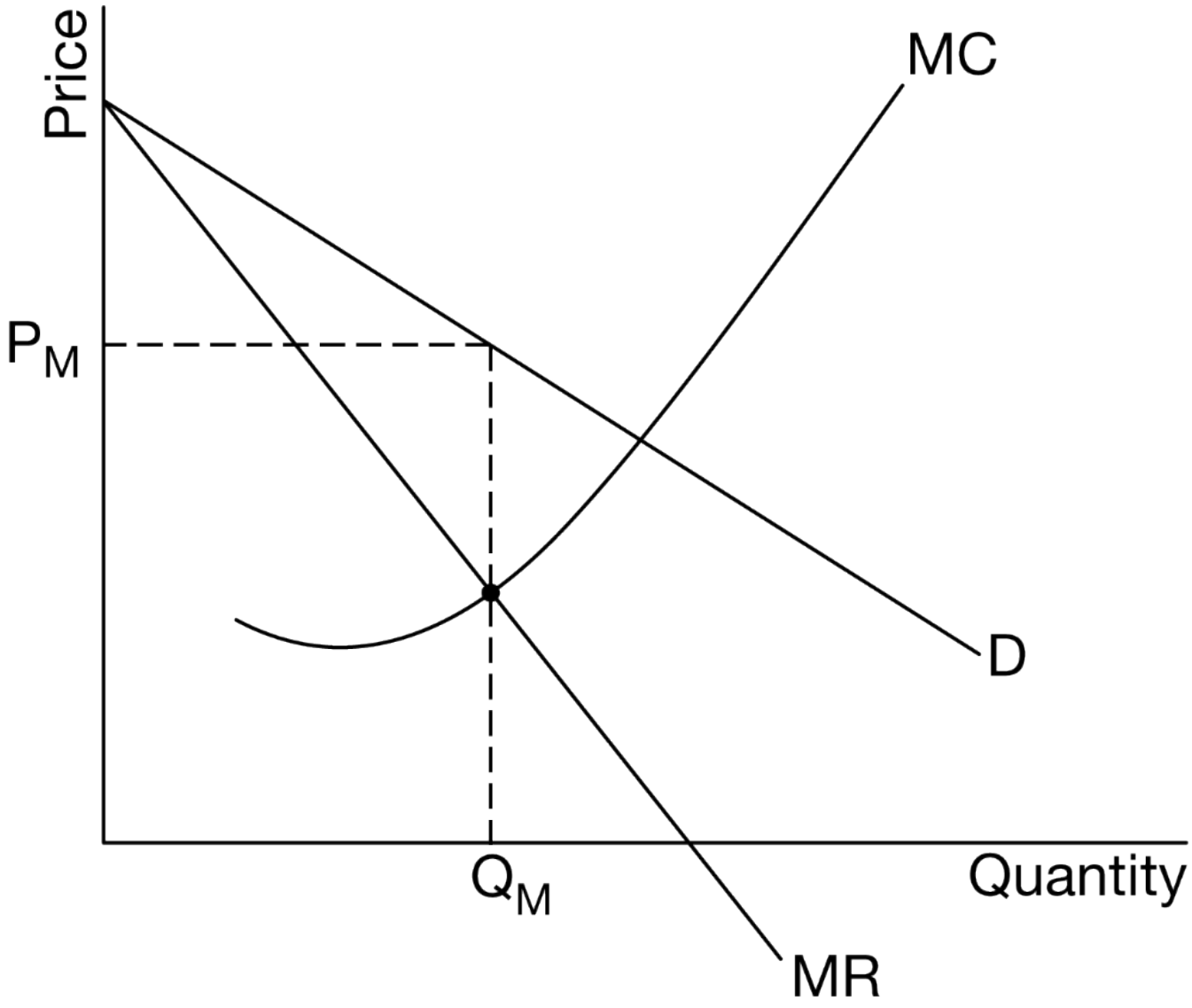
| 0 | 1 | 2 | 3 ✓ |
|---|---|---|---|

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The response accurately includes two of the criteria below.

- ☐ A correctly labeled graph of the firm, with a downward-sloping demand curve and a downward-sloping marginal revenue curve below it.
- ☐ The profit-maximizing quantity labeled $Qm$ at $MR = MC$.
- ☐ $Pm$ from the demand curve above $Qm$.

Solution



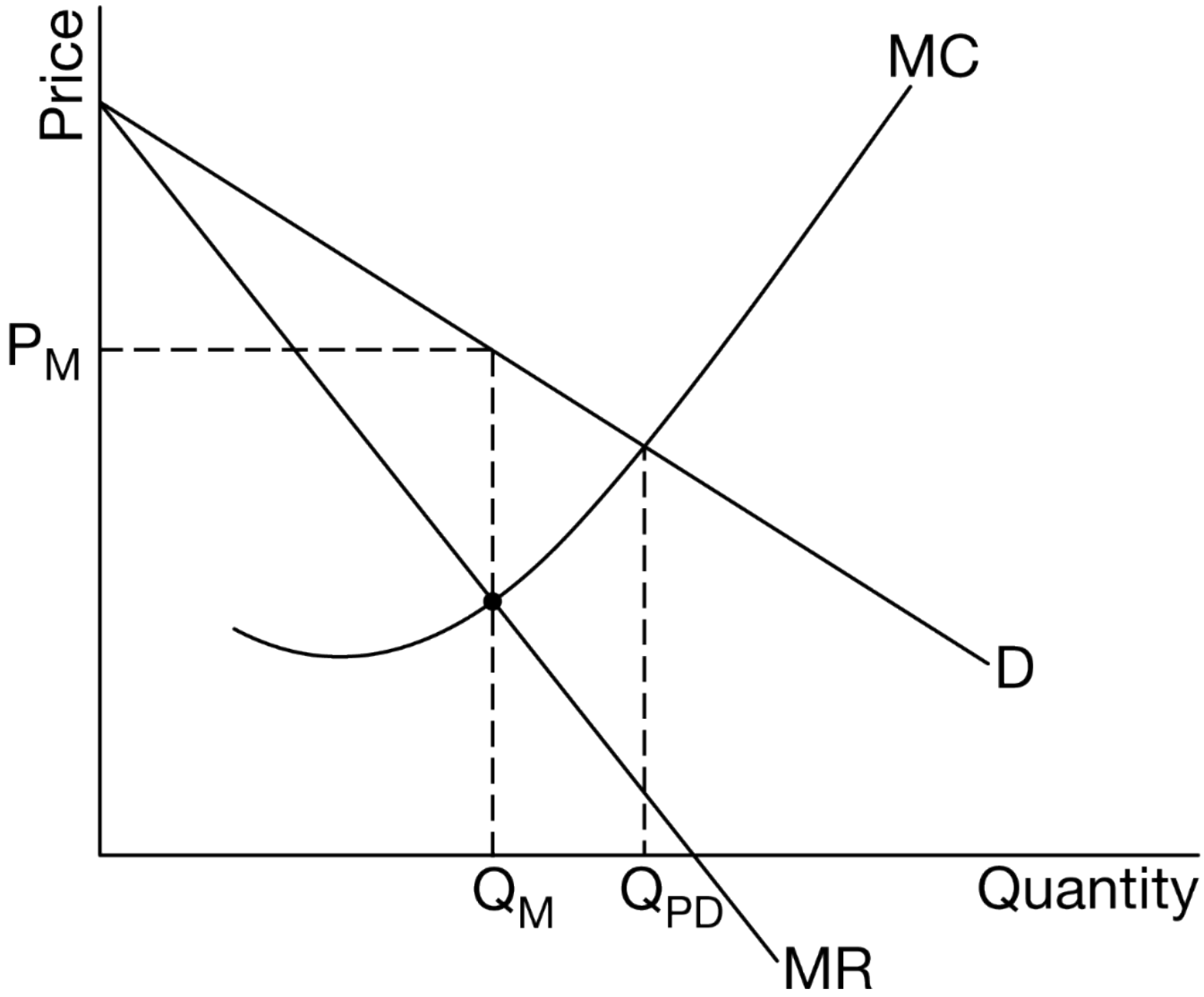

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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| 0 | 1 ✓ |
|---|---|

The response correctly identifies the price discriminating quantity, labeled $Q_{PD}$




The quantity that a price discriminating firm would produce is $Q_{PD}$

**Part (c)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must state a perfectly competitive market will produce the same quantity as the perfectly price-discriminating monopolist and explain why in order to receive the point.

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| 0 | 1 ✓ |
|---|---|

The response states that a perfectly competitive market will produce the same quantity as the perfectly price-discriminating monopolist and explains that perfect competition produces the quantity where $P = MC$ and the perfectly price-discriminating monopolist will produce the same quantity to earn all profit possible.